

Research Article

Chat, Tap, Scan, Pay, and Repeat: WeChat Pay's Continuous Usage Intention Through the S-O-R Model

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China's digital payment landscape is dominated by WeChat (WC), a multifunction social networking platform imbued with a third-party payment application. This study explores the factors that drive the intention to continuously use WC Pay. We propose a framework underpinned by the stimulus-organism-response (S-O-R) model to examine how WC interactivity and consumer-to-consumer interaction influence WC-enabled behavioral engagement and subsequent continuance use intention through perceived value and perceived risk dimensions. Additionally, the study examines personal innovativeness and anxiety as potential moderating factors influencing the link between WC-enabled behavioral engagement and its precursors. Data from 407 Malaysian consumers was gathered using an online survey and analyzed using partial least squares structural equation modeling (PLS-SEM) and moderation testing. The results demonstrate that WC interactivity and consumer-to-consumer interaction indirectly enhance WC-enabled behavioral engagement and the continuance of the platform's use intention through perceived functional and social value. The risk of information exposure does not influence WC-enabled behavioral engagement. In addition, personal innovativeness and anxiety do not serve as moderating variables in the proposed research model. This research contributes to the social media and mobile payment literature by offering a deeper understanding of the drivers and moderating factors affecting the sustained use of WC Pay in Malaysia. Insights derived from this study's findings can inform service providers, marketers, and policymakers to craft practical strategies for bolstering WC Pay's continuous adoption.

Keywords: personal anxiety; personal innovativeness; social media; S-O-R model; WeChat pay

1. Introduction

Mobile payment services have become a staple in modern society thanks to widespread mobile technology and users' demand for secure, safe, and economical transactions [1–3]. WeChat, a versatile social media platform, featured WeChat Pay as its digital wallet tool, which became available to consumers in 2014 [4]. WeChat Pay's significant growth

and popularity among Chinese consumers have prompted its parent company, Tencent, to pursue the platform's global expansion [5]. WeChat is at the forefront of creating opportunities for new business ventures, a trend mirrored by the convergence of commerce and contemporary social networking activities associated with Facebook and Instagram [1, 2]. Increasingly, e-businesses and e-tailers are leveraging mobile payment transactions. WeChat, China's most popu-

lar social media platform, is deeply embedded in users' daily lives, exhibiting significant integration and transaction rates [4].

In the Malaysian context, there is an acceleration of mobile payment adoption [6, 7]. Although there are many mobile payment platforms such as Apple Pay, Grab Pay, Atome, WeChat Pay, Touch n Go, Pay Pal, Ali Pay, and Boost Wallet, WeChat Pay was discontinued in the Malaysian market on September 1, 2024 [8]; however, WeChat Pay users can still use the app to pay where it is supported, but it is not as prominent as other mobile payments, and it left their users in an uncertain situation, thus, making it a platform that is unique to be explored as this study is aimed at examining the antecedents of the continuous usage intentions of WeChat Pay.

Understanding the underlying mechanisms of social media payment systems like WeChat Pay is essential for providing insights into the successful development of financial technologies and commercial platforms. Building upon the stimulus-organism-response (S-O-R) model, this research investigates the determinants of the intention to continue using WeChat Pay among Malaysian consumers. Despite high user engagement with these technological artifacts, there is limited research on the continuance use intention of digital interactive platforms [9]. Hence, it is essential to explore continuance use intention as an outcome of consumer engagement. While a significant amount of research has been dedicated to determining the continued use of mobile payment systems, these studies have primarily focused on the functional value and relative advantages of the product, such as time-saving, convenience, and ease of use [10–13]. This scope has led to knowledge gaps concerning values related to social gratifications. To address the current gaps in the literature, we develop an integrative research model that incorporates perceived social value, a dimension associated with social gratification, along with perceived functional value. The inclusion of the perceived value can be best justified that users are more likely to continuously use mobile payment systems if the systems can promote their social connections among their peers, friends, and family [14] for social approval, status-seeking, and social support [1, 2, 15]. Hence, with the discontinuation of WeChat Pay merchants in Malaysia, it is interesting to examine the perceived value (human-related factor) and whether can it able to invoke continuous usage intention. Our primary objective is to examine how these dimensions of perceived value influence behavioral engagement on WC-enabled platforms and subsequently impact the intention of Malaysian consumers to continue using the technology.

The conceptualization of perceived risk differs across existing S-O-R-related literature on mobile payment adoption. Some studies regard perceived risk as unidimensional [1, 2], while others as a multidimensional construct [16, 17]. Our work considers perceived risk as a multidimensional factor within the context of WeChat Pay while accounting for the role of information exposure risk on WC-enabled engagement and subsequent continued usage. The respective literature documents prior literature, innovativeness, and anxiety as driving personal-related technology

acceptance factors [18, 19]. However, the extent to which these factors serve as moderators in the consumers' adoption intention model is poorly understood. Moreover, even though extensive research has explored the effect of product-related factors on technology adoption, the interplay between personal and product-related factors remains underexplored [20]. Framing personal innovativeness and personal anxiety as moderators within the framework comprising perceived value and perceived risk, the two-fold objectives of this research are as follows:

1. To identify the antecedents and outcomes of WC-enabled behavioral intention.
2. To investigate the moderating effects of personal innovativeness and personal anxiety on WC-enabled behavioral engagement and its antecedents.

This paper is structured as follows: Section 2 describes the theoretical background and presents the hypothesized relationships. Section 2.1 outlines the research design, including details on the sampling procedure, data collection process, statistical techniques employed, and the common method variance (CMV) assessment outcomes. Section 3 presents the results of our data analysis, which encompasses partial least squares structural equation modeling (SEM) (PLS-SEM) and moderation tests. Section 4 discusses the implications, limitations, and recommendations of our study's findings.

2. Literature Review

2.1. S-O-R Model. Mehrabian and Russell [21] introduced the S-O-R model, which highlights the influence of the external environment on individuals' mental processes and behaviors. The stimulus includes elements from the external surroundings, while the organism represents individuals' inner states, encompassing cognitive and emotional aspects [22]. Response pertains to the individual's behavioral reactions, which may include behavioral intention, purchase intention, and continuance use intention [23, 24].

The S-O-R model has been widely applied to studying human behavior, particularly consumer behavior [25, 26]. In a recent study, Kumar et al. [27] draw on the S-O-R model to explore the factors driving consumers to use mobile payment. Relatedly, Rao and Ko [28] and Li et al. [1, 2] revealed that environmental factors, such as system features, system quality, and psychological states like perceived value and satisfaction, play a significant role in shaping human behavior. The existing literature has justified the adequacy and reliability of the S-O-R model as the underpinning theory, applying it to understand the impacts of environmental characteristics on the behavioral reactions of consumers [29]. According to Liu et al. [24], the SOR model delivers a sound and logical technique to evaluate the perspective of consumer behavior. For instance, in the context of social commerce, Wu et al. [30] and Liu et al. [24] indicated that the social commerce environment is devoted to triggering specific emotional or mental reactions of consumers, thus enhancing the likelihood of purchase.

Similarly, the stressor-strain-outcome (SSO) model presents that technology-derived stimuli trigger individuals' psychological states, such as stress, anxiety, and depression. Subsequently, human behavior toward using the technology is shaped [31]. Accordingly, we consider it as an alternative theory to the S-O-R model. In the present study, WeChat, as an advanced e-commerce and social media platform, enables consumer interaction and community formation through an interactive interface, which influences their intention to continue using the platform [1, 2]. Hence, the S-O-R model can be deemed appropriate in our research.

The current study postulates that stimulus contributes to consumers' behavior through their psychological activities. Some research has confirmed the perceived value and consumers' engagement as inner psychological states that affect the consumers' intentions [32–34]. Thus, it is reasonable to associate perceived value and behavioral engagement with the organism element within the S-O-R model. In addition, since perceived risk has been shown as an external factor of organisms [35], we consider the risk of information exposure as an external element apart from perceived value.

2.2. Research Model and Hypothesis Development

2.2.1. WeChat Interactivity. Scholars have conceptualized interactivity from four perspectives: (1) a process of information exchange, (2) a technological characteristic, (3) users' experience of using a system, and (4) a combination of the abovescribed angles [36]. As a process, interactivity is deemed as consumer-to-advertiser interaction [37]. As a characteristic, interactivity concentrates on features of technologies and websites that enable reciprocal communication and user control [37]. Most researchers focus on the third perspective because emphasizing consumers' technology experience is comparatively more important than determining the platform characteristics [36]. In this work, we define interactivity as the extent to which WeChat, as a communication technology, creates an environment that enables real-time reciprocal communication and information exchange.

Existing research has shown an association between interactivity and perceived value [23, 38]. For instance, Lin et al. [38] explored the antecedents leading to the perceived value of food technology in the context of social commerce. The findings advocate interactivity as a driving factor that increases consumers' perceptions of functional benefits. Besides that, Nandi et al. [39] and Kim et al. [32] note that interactivity, as one of the features of mobile applications, enables users to perceive that such apps can meet their needs effectively and conveniently.

Regarding the role of interactivity on social value, as evidenced by Kim et al. [32], social value was insignificantly affected by the interactivity of social media. On the contrary, Hsiao et al. [40] posit that sharing the experience of using social networks creates opportunities to build a typical interaction basis for members of the same social app. Group social connections enable members to recommend an innovative product to others. Li et al. [1, 2] and Tian et al. [15] observe that when an increasing number of WeChat users in an individual's social network use such an innovative pay-

ment method and share favorable experiences, the individual may believe that they will obtain equivalent benefits from the services.

We propose that WeChat interactivity will also relate to consumers' information exposure risk. Szabo and Webster's [41] study addressed the role of perceived interactivity within the green marketing context and found that perceived interactivity negatively affects perceived risk. Meanwhile, the influence of interactivity on risk perception was revealed in Kim's [42] work, illustrating that interactivity can reduce perceived risk. This could be attributed to product-based interactions that can leave a deep impression on the memory of an individual and are conducive to enhancing user satisfaction [42].

The interactivity in virtual platforms can decisively shape the behavior of individuals. According to Liu et al. [24], interactivity, as a characteristic of social commerce, motivates consumers to acquire product-related information, share experiences, and spread information based on real-time chat functions at any time. Consequently, the interactivity encourages consumer-to-consumer interaction. Hajli and Sims [43] also asserted that social media enables consumers to build social ties and foster collaborations. In this regard, interactivity can amplify consumers' proclivity to interact with others via social networks. Thus,

H1: WeChat interactivity positively affects perceived functional value.

H2: WeChat interactivity positively affects perceived social value.

H3: WeChat interactivity negatively affects the perceived risk of information exposure.

H4: WeChat interactivity positively affects consumer-to-consumer interaction.

2.2.2. Consumer-to-Consumer Interaction. Interaction defines the extent to which two or more communication entities can engage with each other in the communication medium and messages [44]. Interaction is a crucial antecedent for successful communication, web design, and electronic commerce in computer-mediated communication [45]. Kim and Tang [46] state that there is a positive association between personal interaction as a form of customer participation and perceived value. This can be explained by the fact that such consumer engagement influences individuals' perception of the value of the product or service at the stage of prepurchase, selection, and acquirement, as well as the stage of usage, possession, and maintenance [46, 47]. Similarly, Lin et al. [38] and Fu and Hsu [48] concluded that consumers tend to acquire valuable information from other consumers or provide others with advice when engaging in social commerce. Liu et al. [24] in their study demonstrate that consumers can perceive more functional value with a higher frequency of communicating with other consumers about the product.

Liu et al. [24] and Yin et al. [49] reveal that interpersonal interactions prompt consumers to build friendships, maintain intimacy, and obtain support from other consumers, thereby inducing a pleasant and satisfactory experience. Interactions among consumers evoke perceptions of mutual empathy and understanding concerning issues, which

contributes to a more solid social identity within a community. Likewise, consumers can derive social value from interacting with others on a social media mobile payment platform [50].

H5: Consumer-to-consumer interaction positively impacts the perceived functional value of consumers.

H6: Consumer-to-consumer interaction positively impacts the perceived social value of consumers.

H7: Consumer-to-consumer interaction negatively impacts the perceived risk of consumers' information exposure.

2.2.3. Perceived Functional Value. This study clarifies WC-enabled behavioral engagement through the lens of perceived functional value. Within this framework, there is ample evidence regarding the connection between perceived functional value and behavioral engagement. For instance, Yu and Zheng [51] aver that consumers can perceive functional value in luxury shopping, which elevates positive attitudes toward products. Similarly, Zhou and Huang [34] and Fu and Hsu [48] revealed that perceived function value enhances the propensity of consumers' shopping behavior. Lai [52] echoes this observation, demonstrating that the value of flipped learning is beneficial to promoting students' behavioral engagement and sustaining effort in their learning. The preceding reflects the extensive literature validating the significant role of perceived functional value. Therefore,

H8: Perceived functional value positively influences the WC-behavioral engagement of consumers to use WeChat Pay.

2.2.4. Perceived Social Value. Perceived social value reflects the product or service's capability to enhance the social image of the customer [53]. Social value pertains to the linkage of a capable product or service with a particular social group. Sheth et al. [54], which prior research has recognized as a noteworthy driver of behavioral engagement. Santos-Vijande et al. [55] assert that the social value individuals acquire from mobile health applications is essential for promoting user engagement. Acquiring a positive social image via mobile health applications can attain a specific level of support, influence, and recognition, thereby promoting a more positive social experience [55]. Likewise, Yu and Zhang [56] and Li et al. [1, 2] indicate that individuals' value perception positively affects engagement in live-streaming shopping. Qian et al. [57] and Pereira et al. [58] acknowledge that individuals' engagement in live-streaming services and omnichannel retail can be augmented by their value perception. We, thus, predict that

H9: Perceived social value positively influences the WC-behavioral engagement of consumers to use WeChat Pay.

2.2.5. Risk of Information Exposure. The risk of information exposure, as an inhibiting factor, is proposed to anticipate behavioral engagement. Perceived risk is defined as consumers' perception of uncertainty and the probable detrimental consequences of purchasing a product [59]. Mer and Virdi [60] and Farzin et al. [61] regard perceived risk as consumers' perceptions of the likelihood and severity of potential adverse consequences of adopting innovative technology. In this research, information exposure risk indicates

individuals' perceptions of how personal information could be misused or lost [62, 63]. According to Khan and Abideen [64], individuals' willingness to use an e-wallet diminishes if they consider it risky. Tripathi [65] observes that consumers' concerns about security and privacy enhance their perception of risk postadoption of mobile wallets. For social commerce, Xue et al. [44] and Tseng [66] show that perceived risk negatively influences consumers' engagement in social commerce, subsequently dissuading them from utilizing such technology. Thus,

H10: The risk of information exposure negatively affects the WC-enabled behavioral engagement of consumers to use WeChat Pay.

2.2.6. WC-Enabled Behavioral Engagement. Ma et al. [67] define behavioral engagement as the extent to which consumers willingly devote their energy, effort, and time to brand-related activities. This study defines WC-enabled behavioral engagement as consumers' active involvement, which entails time, effort, and energy commitment to WeChat Payment activities. Extensive research has supported a significant relationship between behavioral engagement and individuals' intentions [68–71]. Al-Haddad et al. [68] demonstrate a positive correlation between consumer engagement on social media and purchase intention. This finding is corroborated by Guo et al. [69, 70], who identified customer engagement as a significant factor influencing purchase intention, thus highlighting that a high level of engagement contributes to consumer behavior. Zhang et al. [71] found empirical evidence that user engagement positively influences the intention to use mobile payment systems continuously. Thus,

H11: WC-enabled behavioral engagement positively impacts the continuous use intention of WeChat Pay.

2.2.7. Personal Innovativeness. Innovativeness, when considered as a technological factor (technological innovativeness), has been explored as a moderator in extant research [72, 73]. However, its role as a personal factor in influencing consumers' engagement with technology remains relatively unexplored. Personal innovativeness expresses an individual's inclination to seek novel and distinctive experiences [74, 75]. Empirical research conducted by Jiang et al. [76], aimed at validating the moderating role of personal innovativeness, suggests that consumers' innovativeness positively influences the relationship between perceived value and willingness to engage with intellectual property films. Conversely, for those with lower levels of innovativeness, a certain degree of risk is associated with accepting a new product [77]. Consequently, highly innovative individuals tend to embrace challenges, welcome new ideas, and make decisions to evaluate new products [78]. Anwar et al. [79] also demonstrated that consumers' innovativeness can strengthen the relationship between perceived value and their actual engagement in mobile commerce usage. Therefore,

H12: Personal innovativeness moderates the relationships between (a) perceived functional value and (b) perceived social value WC-enabled behavioral engagement.

Specifically, the effect of perceived functional value and perceived social value on WC-enabled behavioral engagement is stronger when consumers' innovativeness is high.

2.2.8. Personal Anxiety. The respective literature has explored how anxiety serves as both a predictive factor and a moderator in adopting specific technologies [80–82]. Yang et al. [82] argue that cognitive anxiety negatively influences students' behavioral engagement in their learning. Additionally, White et al. [83] show the significant impact of anxiety on individuals' willingness to adopt electric vehicles. Hijazi and Abudaabes [81] examine the moderating role of anxiety in the context of mobile payment. Through the lens of prior studies, anxiety can potentially influence individuals' affection, cognitive processing, and decision-making processes when it comes to technology adoption, either strengthening or weakening the relationships between behavioral engagement and its underlying mechanisms. While previous literature has enriched our understanding of anxiety as a driver of behavioral engagement [84], its role as a moderator has been somewhat overlooked. To address this gap, this study is aimed at uncovering the impact of personal anxiety on the relationship between the risk of information exposure and behavioral engagement.

As Zhang [85] observed, negative emotions can significantly impact individuals' motivation, perception, cognition, and behavior. Anxiety, categorized as a negative emotion, denotes a reaction to situational events or objects within an individual's environment that are pertinent to their goals or concerns [85]. Individuals experiencing anxiety may perceive an increased risk, primarily accounting for the potential loss of personal information or data, especially when the responsibility for safeguarding this information is unclear [86]. This heightened perception of risk is associated with the activation of anxiety, which, in turn, leads to a more sensitive perception of behavior. Consequently, individuals with anxiety may be less inclined to engage in financial activities within WeChat Payment due to their elevated perception of the risk of sensitive information disclosure. Thus,

H13: Personal anxiety negatively moderates the relationships between the risk of information exposure and WC-enabled behavioral engagement. Specifically, the effect of risk of information exposure on WC-enabled behavioral engagement is stronger when consumers' personal anxiety is high.

The proposed research model is shown in Figure 1.

3. Methodology

3.1. Research Design. The present research employed a quantitative cross-sectional survey methodology. As defined by Apuke [87], quantitative research involves systematically investigating a phenomenon by collecting quantifiable data and applying mathematical, statistical, or algorithmic techniques. In this study, the survey method was deemed suitable as it allowed the researcher to gain insight into participants' perceptions regarding the research [88]. The research population consisted of Malaysian consumers who met the study's criteria of possessing experience using

WeChat Pay. To gather data, the researcher distributed an online survey to eligible participants.

3.2. Sampling Procedure. We adopted purposive sampling to identify suitable respondents who are WeChat Pay users. Following the purposive sampling protocol that entails a nonprobability technique relying on the researcher's discretion [89], we targeted participants aged 25–39, as this demographic is more likely to be mobile phone users [90]. Before distributing the link to target participants, a pilot study was carried out over 2 weeks, examining the clarity and reliability of the survey instruments. Forty respondents who are WeChat Pay users were invited. As a result of the reliability test, Cronbach's alpha of all constructs is higher than 0.7, which indicates that the reliability meets the acceptable criteria [91].

This study employed an a priori analysis technique, as Memon et al. [92] suggested, to determine the suitable sample size. A power analysis was performed as it is a well-established approach and commonly used in social science research. Using G*power software Version 3.1.9.2, it was established that a minimum sample size of 204 was required for the study. In addition, Kock and Hadaya [93] proposed two methods to determine the sample size, which are the inverse square root and the gamma-exponential methods; however, it is argued that the inverse square root method slightly overestimates the minimum sample size. Thus, researchers opt for utilizing the gamma-exponential method [94], and the minimum sample size required is 146 [92]. However, we acquired 407 usable feedback responses, which is deemed statistically adequate for the study.

3.3. Measurements. The survey questionnaire solicited information regarding demographics, WeChat interactivity, perceived functional value, perceived social value, risk of information exposure, WC-enabled behavioral engagement, continuance use intention, personal innovativeness, and personal anxiety. Demographic variables include gender, age, educational level, and marital status. A screening question was asked to determine respondents' eligibility as WeChat Pay users: "Do you use WeChat Pay for financial transactions?" The survey measured the nine constructs via a 5-point Likert scale, with responses ranging from 1 (*strongly disagree*) to 5 (*strongly agree*).

We measured WeChat interactivity using four items adapted from Liew et al. [95] and Huang and Liao [96]. These items assess the extent to which users can establish social relations, engage in product information exchange, and utilize interaction design features to achieve their goals. They include statements such as "I can establish good social relations with other users and become friends." Consumer-to-consumer interaction was assessed with four items adapted from Nambisan and Baron [50]. These items focus on the frequency of interaction with other consumers on the platform, the helpfulness of the information obtained from the platform, and the sharing of product information with other platform users.

Perceived functional value was assessed via four items, adapted from Sweeney and Soutar [97], containing "Interacting

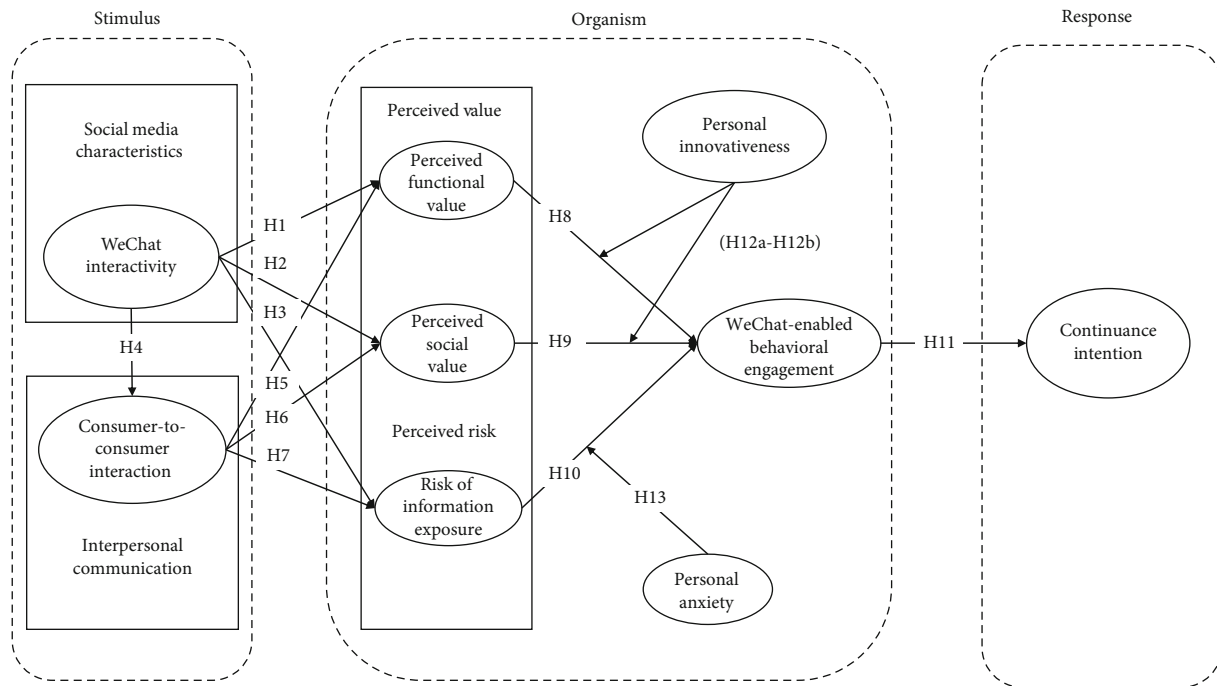


FIGURE 1: Research model.

with other consumers of WeChat Pay can let me know the quality of products/services,” “Interacting with other consumers of WeChat Pay can help me know whether the producing of products/services followed an acceptable standard of quality or not,” and “Interacting with other consumers of WeChat Pay can help me know whether the using of products/services consistent with my initial expectations or not.” Perceived social value is measured by four items, adapted from Sweeney and Soutar [97]. The measurements include “I make a good impression on other people through interacting with other consumers of WeChat Pay,” “I win the admiration of other people through interacting with other consumers of WeChat Pay,” and “I gain approval of other people through interacting with other consumers of WeChat Pay.”

The risk of information exposure was examined via four items, adapted from Johnson et al. [98], Brown et al. [99], and Tan and Teo [100]. The items consist of “I would not feel safe providing personal private information over WeChat Pay,” “I think using WeChat Pay for conducting transactions would have a potential risk, and “Information concerning my transactions via WeChat Pay would be known to others.” Behavioral engagement is examined by four measurements, adapted from Bergkvist and Bech-Larsen [101], including “I often follow news about my WeChat Pay service provider,” “I often talk about my WeChat Pay service provider to others,” and “I often visit my WeChat Pay service providers website or social media.” Continuance use intention was assessed through three items, adapted from Puriwat and Tripopsakul [102] consisting of “I intend to continue using WeChat Pay rather than discontinue their use” and “Continuing to use WeChat Pay for paying goods and services is something I would do.”

Personal innovativeness was measured via four items, adapted from Agarwal and Prasad [103], comprising of “I

heard about WeChat Pay, and I would look for ways to experiment with it,” “Among my peers, I am the first one to try out WeChat Pay,” and “In general, I am not hesitant to try out WeChat Pay.” Personal anxiety is measured by four items, adapted from Rana et al. [104], containing “I feel nervous about using WeChat Pay,” “It scares me to think that I can lose personal information by wrongly using WeChat Pay,” and “I hesitate to use WeChat Pay in fear of making mistakes I cannot correct.”

3.4. Data Collection Procedures. Data for this study were collected through an online survey conducted on various social networking sites such as WeChat, WhatsApp, Instagram, and Telegram. Before completing the survey, participants indicated informed consent, which encompasses assurance that their privacy would be kept confidential and anonymous. Data collection occurred between September 2023 and November 2023, resulting in 407 valid responses after filtering.

3.5. Statistical Technique. Statistical analysis in this study was conducted using PLS-SEM with the assistance of SPSS 26 and SmartPLS 4. The choice of PLS-SEM was motivated by its suitability for analyzing complex research models involving many latent variables. The model comprises 35 items in the present research, making it a complex analytical task.

3.6. CMV. To assess the CMV, a full collinearity assessment was carried out, testing the variance inflation factor (VIF) of the constructs. According to Kock’s [105] criterion, the VIF value should not exceed the threshold of 5. As seen in Table 1, the VIF value of all constructs is less than 5, which concluded that CMV is not a crucial problem in the study.

TABLE 1: Outcome of full collinearity test.

Constructs	VIF
WeChat interactivity	1.734
Consumer-to-consumer interaction	1.719
Perceived functional value	3.894
Perceived social value	3.965
Risk of information exposure	1.742
WC-enabled behavioral engagement	1.777
Personal innovativeness	2.672
Personal anxiety	2.169

TABLE 2: Demographic information ($n = 407$).

Variables	Items	Frequency	Percentage
Gender	Male	196	48.2
	Female	211	51.8
Age (years old)	25 and below	82	20.1
	25–29	190	46.7
	30–34	80	19.7
	35–39	34	8.4
	Above 39	21	5.2
Marital status	Single	180	44.2
	Married	212	52.1
	Divorced	15	3.7
Educational level	Diploma	119	29.2
	Bachelor	191	46.9
	Master	22	5.4
	PhD	9	2.2
	Others	66	16.2

4. Results

4.1. Descriptive Analysis. The demographic characteristics of the 407 participants are presented in Table 2. There is relative equality in terms of gender, with (females comprising 51.8% and males having 48.2%). For age, nearly half fall within the 25–29 years old ($n = 190$, 46.7%). Regarding marital status, more than half of the respondents indicated that they were married ($n = 212$, 52.1%), and nearly half of the respondents have attained a bachelor's degree ($n = 191$, 46.9%).

4.2. Measurement Model Test. Using SEM as the analytical technique, the measurement model underwent evaluation before hypothesis testing. Confirmatory factor analysis (CFA) was employed to assess the measurement model, focusing on the reliability and validity of the nine constructs. For assessing reliability, two indicators, Cronbach's alpha and factor loadings, were considered to determine the internal consistency of the variables. Table 3 presents the results of the reliability test. Following the criteria of Hair et al. [91], Cronbach's alpha values should exceed 0.7, while factor loadings should be greater than 0.5. As shown in Table 3,

all constructs in this study met or exceeded these threshold values, indicating an acceptable level of reliability.

The validity assessment comprises convergent validity and discriminant validity. Convergent validity was evaluated using composite reliability (CR) and average variance extracted (AVE), with threshold values of 0.7 and 0.5, respectively [106]. Based on the CFA results, all constructs in this study exceeded these threshold values for both CR and AVE, indicating acceptable convergent validity. Discriminant validity was assessed using the heterotrait–monotrait (HTMT) ratio, as displayed in Table 4. Following the criteria proposed by Henseler et al. [107] and Ma et al. [67], discriminant validity is considered acceptable if the HTMT index between constructs does not exceed 0.90. Table 4 demonstrates that all HTMT values are below 1, indicating acceptable discriminant validity.

4.3. Structural Model Assessment. Pathway analysis was conducted using PLS-bootstrapping to assess the structural model, and the study's findings are presented in Table 5 and Figure 2. As anticipated, the results demonstrated a positive relationship between WeChat interactivity and perceived functional value (Beta = 0.356, $T = 7.676$, $p < 0.05$), affirming H1. Similarly, WeChat interactivity positively influences perceived social value (Beta = 0.374, $T = 7.742$, $p < 0.05$), validating H2. Consistent with our hypothesis, WeChat interactivity negatively impacts the risk of information exposure (Beta = 0.276, $T = 4.742$, $p < 0.05$), supporting H3. The findings also confirmed a significant connection between WeChat interactivity and consumer-to-consumer interaction (Beta = 0.582, $T = 15.978$, $p < 0.05$), thus validating H4.

In line with our hypothesis, consumer-to-consumer interaction positively correlates with perceived functional value, endorsing H5. The data also indicates that consumer-to-consumer interaction positively affects perceived social value (Beta = 0.327, $T = 6.925$, $p < 0.05$), thereby affirming H6. Furthermore, there is a negative relationship between consumer-to-consumer interaction and the risk of information exposure (Beta = 0.123, $T = 2.19$, $p < 0.05$); thus, H7 was supported.

Another significant relationship was identified between perceived functional value and WeChat-enabled behavioral engagement (Beta = 0.179, $T = 2.262$, $p < 0.05$), confirming H8. Moreover, the outcome of the pathway analysis revealed that perceived social value significantly affects WeChat-enabled behavioral engagement (Beta = 0.193, $T = 2.578$, $p < 0.05$), validating H9. Contrariwise, an insignificant association was found between the risk of information exposure and WeChat-enabled engagement (Beta = -0.099 , $T = 1.717$, $p > 0.05$), leading to the rejection of H10. Lastly, the strongest positive correlation was observed between WeChat-enabled engagement and continuance intention (Beta = 0.629, $T = 19.317$, $p < 0.05$), endorsing H11.

4.4. Moderation Test. Based on Table 6 and Figure 2, the results of the moderation test indicated that personal innovativeness has no significant effect on the relationship between perceived functional value (Beta = -0.013 , $T =$

TABLE 3: Convergent validity.

Construct	Items	Loadings	Cronbach's alpha	CR	AVE
WeChat interactivity (WI)	WI1	0.652	0.705	0.727	0.526
	WI2	0.754			
	WI3	0.708			
	WI4	0.780			
Consumer-to-consumer interaction (CTC)	CTC1	0.653	0.726	0.738	0.547
	CTC2	0.813			
	CTC3	0.733			
	CTC4	0.751			
Perceived functional value (PFV)	PFV1	0.892	0.906	0.908	0.781
	PFV2	0.892			
	PFV3	0.876			
	PFV4	0.874			
Perceived social value (PSV)	PSV1	0.740	0.883	0.904	0.742
	PSV2	0.895			
	PSV3	0.915			
	PSV4	0.884			
Risk of information exposure (RIE)	RIE1	0.842	0.844	0.853	0.68
	RIE2	0.840			
	RIE3	0.790			
	RIE4	0.826			
WC-enabled behavioral engagement (BE)	BE1	0.790	0.862	0.87	0.707
	BE2	0.845			
	BE3	0.862			
	BE4	0.863			
Continuance intention (CI)	CI1	0.859	0.758	0.819	0.664
	CI2	0.713			
	CI3	0.863			
Personal innovativeness (PI)	PI1	0.766	0.826	0.835	0.656
	PI2	0.842			
	PI3	0.806			
	PI4	0.823			
Personal anxiety (PA)	PA1	0.855	0.791	0.828	0.613
	PA2	0.635			
	A3	0.790			
	PA4	0.833			

0.124, LLCI = -0.196, ULCI = 0.201) and perceived social value (Beta = 0.07, $T = 0.704$, LLCI = -0.141, ULCI = 0.246) with WeChat-enabled behavioral engagement. This means that even when personal innovativeness is high, the influence of perceived functional value and perceived social value on WeChat-enabled behavioral engagement remains unchanged. Therefore, H12a and H12b are rejected. Similarly, personal anxiety was found to have no significant effect on the relationship between the risk of information exposure and WeChat-enabled behavioral engagement (Beta = 0.078, $T = 1.674$, LLCI = -0.006, ULCI = 0.176). Consequently, H13 is also rejected.

5. Discussion and Conclusions

Grounded on the S-O-R model, we aimed to uncover the factors influencing Malaysian consumers' intention to use WeChat Pay continuously. By examining perceived value and perceived risk, a comprehensive model was articulated to understand the drivers of consumers' intention to continue using WeChat Pay through their engagement with the platform. Additionally, this study investigated how personal innovativeness and personal anxiety might moderate the impact of perceived value and perceived risk on consumers' behavior.

TABLE 4: Heterotrait–monotrait ratio.

	WI	CTC	PFV	PSV	RIE	BE	CI	PI	PA
WI									
CTC	0.740								
PFV	0.664	0.652							
PSV	0.699	0.663	0.840						
RIE	0.455	0.338	0.472	0.503					
BE	0.513	0.649	0.636	0.624	0.288				
CI	0.649	0.607	0.787	0.779	0.539	0.722			
PI	0.680	0.596	0.761	0.783	0.776	0.528	0.810		
PA	0.589	0.583	0.758	0.767	0.569	0.677	0.831	0.731	

The pathway analysis revealed that WeChat interactivity significantly impacts perceived functional value. This finding echoes those in Lin et al.'s [38] work, which also explored the relationship between WeChat interactivity and perceived functional value. Lin et al. [38] similarly found that social commerce interactivity enhances consumers' perception regarding the functional value of WeChat. Furthermore, the study confirmed that WeChat interactivity plays a crucial role as an antecedent of perceived social value. This finding is consistent with the research conducted by Kim et al. [32] and Li et al. [1, 2], demonstrating the positive impact of social interactivity on users' perceived social value. These studies suggest that social media interactivity facilitates establishing social connections and sharing experiences, leading consumers to believe they can benefit from such services.

Additionally, the study revealed that WeChat interactivity negatively impacts the risk of information exposure. This observation conforms with the findings of Szabo and Webster [41] and Kim et al. [42], who concluded that interactions related to products or services can reduce perceived risk. Based on the perspective of Song and Liu [108], interactivity is found to have no relationship with perceived risk in the context of live-streaming, which can be explained that consumers may hide unwanted and unwelcome information about the transaction if they perceive streamers as a source of risk. Collectively, this implies that consumers' interactions on WeChat contribute to alleviating perceptions of risk regarding the exposure of their personal information and privacy.

The SEM analysis showed that WeChat interactivity positively affects consumer-to-consumer interaction. This outcome is congruent with Liu et al. [24] and Hajli and Sims [43], who revealed a positive relationship between interactivity and consumer interaction. According to Liu et al. [24], interactivity enables consumer-to-consumer interaction. Similarly, Hajli and Sims [43] evidenced that social network interactivity can encourage consumers to interact with others and create social connections. Our result, thus, indicated that WeChat, with the characteristic of interactivity, encourages consumers to communicate with others about the product.

The findings of this study support the hypothesis suggesting a significant impact of consumer-to-consumer inter-

action on perceived functional value. This observation aligns with Kim and Tang's [46] study, which demonstrated a significant relationship between personal interaction and perceived value. Likewise, Liu et al. [24] revealed that consumers tend to believe that they can derive more functional benefits when engaging in product-related interactions with other consumers. These findings signify that consumers who use WeChat as an interactive platform to engage with others can enhance their perceptions of the functional benefits offered by WeChat Pay.

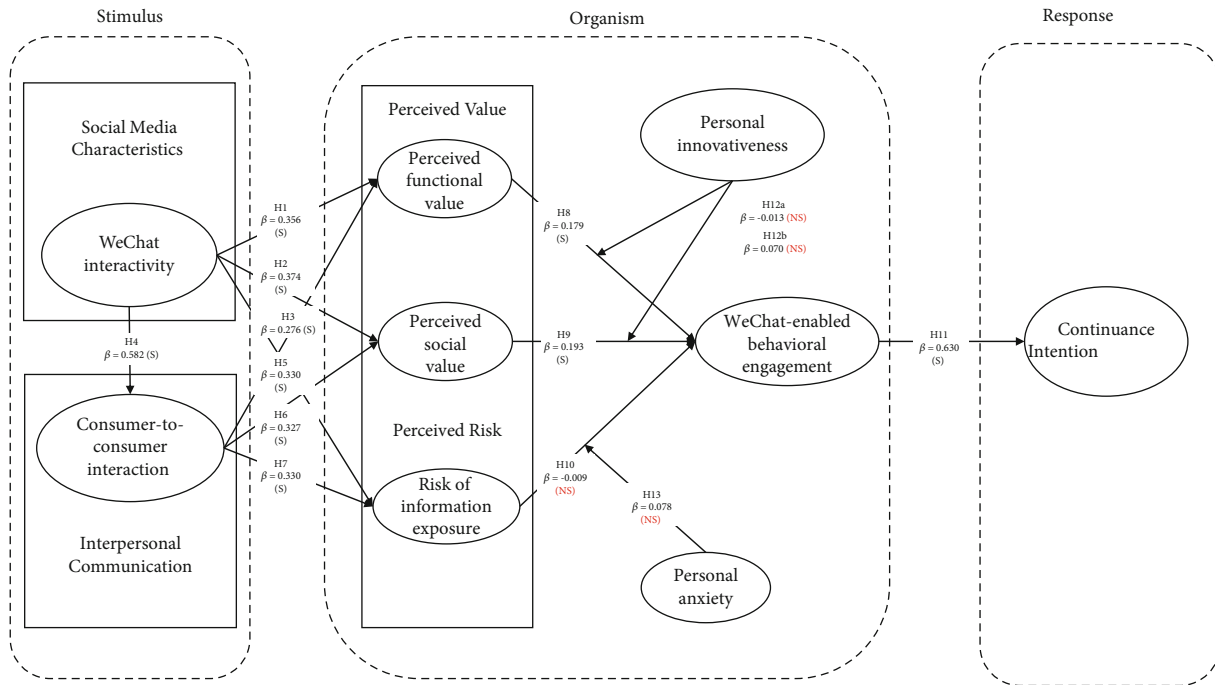
Consumer-to-consumer interaction also proves to be a crucial predictor of perceived social value, thereby supporting the proposed hypothesis. Our study aligns with the conclusions of Yin et al. [49], who highlighted the impact of consumers' social interactions. This finding in our research is also congruent with Maryanto et al. [109] who illustrated that parasocial interactions between social influencers and consumers trigger the consumers' emotional reactions, constructing parasocial relationships. Consequently, individuals may derive social trust and great impressions toward the products. However, our research goes a step further by providing additional insights. We demonstrate that the degree and intensity of interaction among Malaysian consumers on WeChat are positively associated with their belief that their social image can be improved through its usage. In simpler terms, the more Malaysian consumers interact on WeChat, the more they perceive that their social image can benefit from it. In essence, the information, insights, and recommendations exchanged during these interactions lead to an improved recognition of WeChat's social benefits.

The connection between consumer-to-consumer interaction and the risk of information exposure is confirmed, aligning with the findings of Xue et al. [44], who observed a negative impact of interaction on consumers' risk perception in the context of social commerce. This result is also consistent with the viewpoint of Liu et al. [24], who provided evidence of the adverse effect of consumer-to-consumer interaction. Besides, as Tsoy et al. [110] mentioned, the acquisition and transmission of information through social media can shape individuals' risk perception by amplifying their fears and sense of uncertainty. For instance, during the period of COVID-19, the intensity of feeling fear might be augmented with an increasing level of information obtained from social media, which delivers a consistent viewpoint to the present research. In essence, our findings suggest that consumer-to-consumer interactions can enhance users' familiarity and understanding of the WeChat system, thereby reducing their sensitivity to potential risks.

In line with the previous research by Zhou and Huang [34], who emphasized the role of perceived functional value in influencing consumers' shopping behavior, we have validated the impact of perceived functional value on WC-enabled behavioral engagement. Drawing from Lai's perspective [52] that students' engagement in flipped learning can be enhanced when they perceive the utility value in their learning, our findings suggest that Malaysian consumers are more likely to engage with WeChat Pay when they perceive a certain level of functional value.

TABLE 5: Result of hypothesis testing (one-tailed test).

	Hypotheses	Std. beta	Std. errors	T values	Bias corrected	LLCI (5%)	ULCL (95%)	Decision	VIF	R ²	Q ²	f ²
H1	WI→PFV	0.356	0.046	7.676	0.000	0.279	0.431	Support	1.512	0.373	0.286	0.134
H2	WI→PSV	0.374	0.048	7.742	0.001	0.287	0.447	Support	1.512	0.389	0.284	0.151
H3	WI→RIE	0.276	0.058	4.742	0.003	0.172	0.366	Support	1.512	0.131	0.084	0.058
H4	WI→CTC	0.582	0.036	15.978	0.002	0.516	0.636	Support	1.000	0.339	0.175	0.512
H5	CTC→PFV	0.330	0.051	6.506	0.002	0.245	0.412	Support	1.512			0.115
H6	CTC→PSV	0.327	0.047	6.925	0.001	0.246	0.400	Support	1.512			0.116
H7	CTC→RIE	0.123	0.056	2.190	0.000	0.031	0.215	Support	1.512			0.012
H8	PFV→BE	0.179	0.079	2.262	0.001	0.047	0.308	Support	4.157	0.432	0.288	0.014
H9	PSV→BE	0.193	0.075	2.578	0.000	0.066	0.314	Support	4.017			0.016
H10	RIE→BE	-0.099	0.058	1.717	-0.003	-0.194	-0.005	Reject	2.061			0.008
H11	BE→CI	0.630	0.033	19.317	0.002	0.569	0.677	Support	1.000	0.397	0.239	0.654



S = Supported; NS = Not Supported

FIGURE 2: Path diagram of the research model.

TABLE 6: Result of moderation testing (one-tailed test).

Hypotheses	Std. beta	T values	Bias correlated	p values	LLCI (5%)	ULCI (95%)	Decision
H12a PI × PFV → BE	-0.013	0.124	0.011	0.902	-0.196	0.201	Rejected
H12b PI × PSV → BE	0.070	0.704	-0.006	0.481	-0.141	0.246	Rejected
H13 PA × RIE → BE	0.078	1.674	-0.007	0.094	-0.006	0.176	Rejected

For example, consumers who perceive value are more inclined to share their knowledge about WeChat Pay, interact with others regarding its benefits, and recommend it to friends [111].

Consistent with the findings of previous studies by Santos-Vijande et al. [55], Tian et al. [15], and Yu and Zhang

[56], which emphasized the influence of perceived social value on WC-enabled behavioral engagement, our study also confirms the positive impact of perceived social value, thereby supporting the hypothesized relationship. Our results show that consumers' perception of social value motivates them to invest time and effort actively in using

WeChat Pay. On the other hand, our findings do not support the relationship between the risk of information exposure and WC-enabled behavioral engagement. This result countervails the findings of Xue et al. [44], who concluded that perceived risk significantly and negatively affects social commerce engagement. It is plausible that consumers place greater emphasis on the advantages of WeChat, such as utility and social benefits, and their concerns about WeChat Pay are mitigated through interpersonal interactions with others.

The data of this study suggests that WC-enabled behavioral engagement drives continuance intention. This finding echoes Al-Haddad et al. [68] and Zhang et al. [71], who indicated that consumer engagement positively relates to the willingness to adopt technology, such as social media and mobile payment systems. Therefore, it can be inferred that by actively using WeChat to maintain social connections with others, Malaysian consumers are more likely to have a stronger intention to continue using the system.

Regarding the moderating effect of personal innovativeness, the findings suggest that personal innovativeness does not impact the relationship between perceived functional value and perceived social value. This result is aligned with the evidence provided by Tian et al. [112], who highlighted the insignificant moderating effect of personal innovativeness in adopting the e-wallet system. For this study, we conclude that Malaysian consumers' innovativeness, curiosity, and intrinsic motivations do not diminish the effect of their perception of the functional and social value of WeChat Pay. Thus, it can be concluded that their personal traits will not strengthen their inclination to use WeChat payment anticipated by the system's advantages. A potential explanation is that users with innovativeness also concentrate on the inherent uncertainty in online payment activities, inclining to mitigate the potential risks [42, 113]. Hence, they are cautious to embrace behavioral engagement despite a high level of curiosity demonstrated.

Finally, the moderating effect of personal anxiety was found to be insignificant, indicating that personal anxiety does not strengthen the impact of the risk of information exposure on WC-enabled engagement. This result conflicts with the findings of Grable et al. [114], who demonstrated a significant moderating role of personal anxiety in the context of financial behavior. Retrospectively, this shows that Malaysian consumers' perception of their privacy exposure on their WC-enabled behavioral engagement is not amplified under high sensitivity and anxiety conditions. According to Zhu et al. [115], individuals with high levels of anxiety are more likely to take effective strategies, mitigating the detrimental effects of anxious symptoms on their mental health. Besides, Almaiah et al. [116] emphasized that individuals' anxiety can be reduced through social interactions through innovative tools. Accordingly, WeChat interactivity serves as a technological factor enabling a diminishing level of personal anxiety, thus not amplifying the association between the risk of information exposure and behavioral engagement. In the current research, WeChat Pay users with anxious symptoms, to reduce their sense of uncertainty, are enabled to perform corresponding actions, such as avoidance behavior and switching behavior as reactions. Hence,

by decreasing the level of personal anxiety, users may not passively engage in the use of WeChat Pay if they perceive using WeChat payment as risky.

5.1. Theoretical Implications. The present research offers three significant contributions. Firstly, it introduces a comprehensive framework to the existing literature on social media and mobile payment. While the S-O-R model has been widely applied in studies related to VR shopping, artificial intelligence, and mobile commerce [25, 117, 118], there is relatively limited research that focuses on adoption behavior in the context of WeChat Pay. Building upon the S-O-R model, this study analyzes how Malaysian consumers process values and risks when deciding to continue using the WeChat Pay system after its operation ceased. The research highlights that perceived value and perceived risk, both as internal states and product-related factors, play crucial roles in shaping consumers' behavioral engagement and continuance use intention. By emphasizing the significance of the S-O-R concept, this study provides valuable theoretical insights into understanding consumer behavior in the context of WeChat Pay postadoption in Malaysia. We frame WeChat interactivity and consumer-to-consumer interaction as stimuli and demonstrate that perceived social value has a more substantial impact than perceived functional value. This expands the perspective on socialization-based influential factors within social media to the mobile payment literature. Consequently, this research helps bridge the gap in prior literature that had limited integration between social media and mobile payment. Our findings contribute to the broader field of consumer behavior by shedding light on how Malaysian consumers perceive and engage with WeChat Pay. It provides valuable insights for businesses and policymakers in the mobile payment industry to understand better and cater to the needs and preferences of consumers in Malaysia.

Secondly, this research enhances understanding of the dimensions of perceived value, including facilitators such as perceived functional value and social value, as well as the dimension of perceived risk (risk of information exposure) as a barrier. Previous studies focused primarily on the role of facilitators when determining usage intention or behavior, with limited attention given to barriers [119, 120]. This study empirically examines the continued usage intention of WeChat Pay from value and risk perspectives, offering a unique perspective on the mechanisms of consumer behavioral engagement in social media and mobile payment. Additionally, Li et al. [1, 2] pointed out that perceived functional value, being a tangible value, has received significant attention in research, while other intangible values, such as perceived interactive value and perceived social value, remain relatively unexplored. This research introduces perceived social value as an intangible value associated with WeChat, investigating its contribution to WC-enabled behavioral engagement and continuance use intention. Consequently, our findings empirically validate the distinctive aspects of WeChat Pay compared to other mobile payment systems, particularly in perceived functional and social value dimensions. This provides a comprehensive

insight into the role of nonfunctional value in enhancing the continued usage of social media payment platforms.

Thirdly, WC-enabled behavioral engagement, as a dimension of behavioral engagement, was explored in the current research. As emphasized by Roy et al. [9], the consumer, an inherently interactive concept, serves as a mechanism to look at consumers' relationship with digital interactive platforms. From the perspective of WeChat, an interactive platform, we examined how WC-enabled engagement is driven by the interaction of personal-related factors and product-related factors. This contributes to the literature on consumer behavior by looking into the complex associations between behavioral engagement, continuance intention, and their predictors. In addition, the conceptualization and dimensionality of consumer engagement have also been a controversial topic [9]. For instance, some researchers noted that consumer engagement contains both in-role and extrarole (cognition, emotion, and behavior) [121]. Yet, another group of researchers identified consumer engagement as a unidimensional construct [122]. In the current study, we conceptualized it as a multidimensional, offering a more comprehensive view of consumer engagement.

Fourthly, this research contributes to understanding the moderating role of personal innovativeness and personal anxiety, which are influential personal-related factors in social media payment. Previous research on social media payment has relatively neglected to examine whether and how these factors can act as moderators between product-related predictors and usage intention [20]. This study unpacked the interaction between product-related and personal factors. Although the moderating effects of personal innovativeness and personal anxiety were not observed, this research addresses the literature gap by accounting for their potential moderating roles. Furthermore, this study extends the S-O-R model and demonstrates the effectiveness of the extended research model. This extension enhances our understanding of the complex interplay between various factors influencing continued usage intention in the context of social media payment.

5.2. Practical Implications. This study's outcomes hold significant practical implications for marketers, service providers, and policymakers, particularly in understanding the dynamics behind the sustained use of WeChat Pay, a social media-integrated mobile payment system, among Malaysian consumers. For marketers and service providers aspiring to penetrate the mobile payment market and enhance WeChat Pay usage, grasping the influence of perceived value on user continuation is vital. Given the shift of numerous Malaysian consumers to mobile payment, leveraging the experience of current WeChat Pay users can boost market penetration. Additionally, to maintain and grow WeChat Pay's user base, service providers should maximize value by integrating more social functions and improving user-centric features like convenience and a humanized interface. In addition, the service providers should also encourage Malaysian Chinese consumers to have a China's number to able to access the

newest features of WeChat Pay, as China has ranked number one in the market of mobile payment [123].

As a result of the relationship between risk of risk of information, despite the lack of significant findings regarding the risk of information exposure, it is expected to benefit system users and service providers. According to Roy et al. [9] and Ou et al. [124], consumer engagement denotes the investment of consumers' resource investments. Therefore, in the context of WeChat Pay, it is called to reduce the sense of uncertainty in terms of the potential loss of consumers' financial information. The findings encourage users of WeChat Pay to engage in proactive communications with marketers, stimulating their sense of safety and system value toward the platform. Apart from that, service providers are recommended to conduct continuous survey research to identify and mitigate barriers deterring the ongoing use of WeChat Pay. To address concerns about privacy and security [125], such as cyber fraud and hacking, policymakers should develop regulations to safeguard user privacy and data. Service providers, in turn, can bolster user confidence by implementing technical safeguards like encryption, facial recognition, and authentication processes.

5.3. Limitations and Future Directions. Although our research has made several substantial contributions, we acknowledge certain limitations that should be taken into account for future investigations. Firstly, this study examined the dimensions of perceived value contributing to WC-enabled behavioral engagement and continued usage of WeChat Pay. It is worth noting that previous research [126, 127] has identified other dimensions of value derived from social applications, including emotional value, social communication value, and information value. Future researchers should explore these additional dimensions to provide a more comprehensive understanding.

Secondly, our examination of the moderation effect of personal innovativeness and personal anxiety was based on Malaysian consumers. To enhance the generalizability of our findings, it would be beneficial for researchers to apply and validate the research model in different countries or regions. This can help uncover potential variations in user behavior and the role of perceived value and risk across diverse cultural and geographic contexts.

Lastly, our study found that the moderation effects of innovativeness and anxiety were statistically insignificant. However, it is essential to acknowledge that these effects may vary in different research contexts or with different samples. Therefore, we recommend that future researchers continue investigating these factors, as their influence might become more apparent under different conditions.

Furthermore, when exploring consumers' reactions to technology, it is crucial to consider that individuals may experience varying anxiety levels. Future studies could benefit from specifying anxiety levels (low, moderate, and high) within the context of social media payment postadoption. This segmentation approach can provide valuable insights into how different levels of anxiety impact users' risk perceptions and engagement with payment platforms. Specifically,

it may reveal that consumers with high anxiety perceive a greater risk of information exposure compared to those with low personal anxiety. Our study's area of inquiry remains relatively underexplored, and future research can contribute to a more nuanced understanding of these dynamics by examining various barriers [128].

Data Availability Statement

The data used to support the findings of this study are available from the first author upon reasonable request.

Consent

Informed consent was obtained from all subjects involved in the study. Respondents were informed through the survey indicated on the voluntary participation and confidentiality of the respondent's information.

Conflicts of Interest

The authors declare no conflicts of interest.

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